

Evaluating Causal Reasoning in Large Language Models through Standardized Datasets and Dual-Agent Explanation Assessment

Sonja Gievska*, Slobodan Kalajdziski*,
Martina Toshevska*, Andrea Stevanoska*,
Viktor Kostadinovski*, Gorazd Filipovski*,
Nikola Jagurinoski*, Boris Smokovski*

* Faculty of Computer Science and Engineering, Ss. Cyril and
Methodius University in Skopje

Abstract

We present a systematic framework for evaluating the causal reasoning abilities of large language models (LLMs). Unlike prior work that often focuses on narrow, domain-specific tasks, our approach unifies datasets from diverse sources—including code, mathematics, text, and entailment—into a standardized evaluation pipeline. The framework supports both prediction and explanation generation, enabling a dual assessment of correctness and interpretability. By incorporating flexible prompt construction, configurable model selection, and detailed tracking of accuracy, explanation quality, execution time, and API cost, the method offers a comprehensive view of LLM performance. Pre-computed outputs and exploratory analysis tools further highlight the practicality of the framework. Empirical results demonstrate that this approach provides scalable, reproducible, and extensible evaluation of causal reasoning, paving the way for more robust benchmarking of LLMs across domains.

1 Introduction

Large language models (LLMs) have rapidly advanced in recent years, demonstrating strong performance across tasks ranging from solving mathematical problems and writing code to generating long stories and answering open-ended questions [1, 2]. These models, trained on massive text corpora, capture both general knowledge and domain-specific patterns. Notably, researchers have observed that LLMs sometimes exhibit behaviors not explicitly taught, such as reasoning about scenarios, explaining their choices, or considering counterfactual situations [5, 6]. This progress naturally raises an important question: can LLMs reason about cause-and-effect relationships?

Causal reasoning is a fundamental aspect of human cognition, enabling understanding of not just correlations but the underlying causes of events [8, 7]. These relationships are often represented through causal graphs, where nodes denote variables and edges indicate the direction of influence. Accurate causal graphs form the basis for predicting outcomes, estimating effects, and understanding the consequences of interventions. Traditional causal discovery methods heavily rely on numerical data and statistical techniques [3, 4]. While effective, these approaches can be challenging when data is limited or noisy. Recent studies suggest that LLMs might offer an alternative by using metadata, such as variable names and descriptions, to infer and explain causal connections [1, 2].

Despite these promising findings, systematic evaluation of LLMs’ causal reasoning remains limited. Most existing work focuses on small, domain-specific problems or pairwise comparisons between variables, which do not scale to larger graphs and provide limited insight into the robustness of causal inference [1, 2]. Without a comprehensive framework, it is difficult to ascertain whether LLMs are genuinely reasoning about causality or merely producing plausible-sounding answers.

In this work, we present a unified evaluation framework that assesses LLMs’ causal reasoning across multiple domains, including mathematics, programming, natural language, and logical entailment. Our framework evaluates both the accuracy of causal predictions and the quality of explanations, offering a more holistic view of performance. It supports flexible prompting, multiple model choices, and detailed logging of factors such as cost and runtime, enhancing transparency and reproducibility. Pre-computed outputs and exploratory notebooks allow researchers to engage with the framework immediately, without re-running expensive experiments. By combining scalability, reproducibility, and extensibility, this framework lays the groundwork for more reliable benchmarking of LLMs’ causal reasoning abilities, emphasizing not only the correctness of answers but also the justification behind them [5, 8, 7].

2 Related Work

Our project builds on recent work at the intersection of causal inference and large language models (LLMs). Classical causal discovery methods, such as the PC algorithm [9], Greedy Equivalence Search (GES) [10], and NOTEARS [11], have traditionally relied on observational data and statistical independence testing. While theoretically well-grounded, these approaches face limitations in scalability, noise sensitivity, and their dependence on large volumes of structured data. The emergence of LLMs has opened a new research direction: using natural language descriptions and domain knowledge to aid causal reasoning.

One line of work that strongly influenced our approach is the paper *Can Large Language Models Build Causal Graphs?* by Long, Schuster, and Piché (2023) [12]. The authors investigated whether LLMs could assist researchers in constructing causal diagrams, with a particular focus on medical contexts. Their experiments revealed that LLMs can sometimes infer plausible causal

edges from text descriptions, but their performance is highly sensitive to prompt design and the specific phrasing of causal relationships (e.g., “causes” versus “increases risk”). They concluded that while LLMs show promise as assistants in causal modeling, expert oversight remains indispensable because the models may overlook confounding variables or misinterpret causal directions. This work inspired our use of standardized datasets and explanation validation, since it highlighted the fragility of relying solely on raw LLM outputs.

Another key influence was the paper *Efficient Causal Graph Discovery Using Large Language Models* by Jiralerspong, Chen, More, Shah, and Bengio (2024) [13]. In contrast to earlier pairwise-query approaches, which scale quadratically with the number of variables, their framework employs a breadth-first search (BFS) strategy that reduces query complexity to linear time. By leveraging the Directed Acyclic Graph (DAG) property and explicitly enforcing acyclicity, their method achieves state-of-the-art results on both small benchmark graphs (Asia and Child) and large-scale graphs such as the Neuropathic Pain dataset with over 200 nodes. They also demonstrated that incorporating observational statistics into prompts further boosts accuracy, bridging textual and numerical methods. This efficiency-oriented approach motivated us to emphasize modularity and cost control in our own pipeline.

Together, these two studies provided the foundation for our work. Long et al. emphasized the potential and limitations of LLMs in causal graph construction, while Jiralerspong et al. demonstrated how LLM-based methods can be scaled up to larger graphs.

3 Solution Architecture

Our system is a modular pipeline for evaluating causal reasoning in large language models (LLMs). It transforms heterogeneous causal datasets into a unified schema, produces model predictions and explanations, assesses explanation quality with a second LLM, and aggregates dataset-level metrics and metadata. The four principal stages are: (i) Dataset Mapping and Standardization, (ii) Causal Reasoning Agent (predictions), (iii) Explanation Evaluation Agent (explanations), and (iv) Evaluation and Reporting. An overview is shown in Figure 1.

3.1 Notation and Problem Setup

Let $\mathcal{D}^{\text{raw}} = \{r_i\}_{i=1}^N$ denote records from a raw dataset with arbitrary column names and formats. The goal is to evaluate an LLM’s ability to answer causal questions and to justify its answers with explanations that are both semantically aligned and causally sound relative to provided references.

We define a user-supplied *DatasetMapping* ϕ which specifies a column-wise correspondence from the raw schema to a standardized schema (Table 1). Applying ϕ yields a standardized dataset $\mathcal{D}^{\text{std}} = \{s_i\}_{i=1}^N$ where each instance s_i is

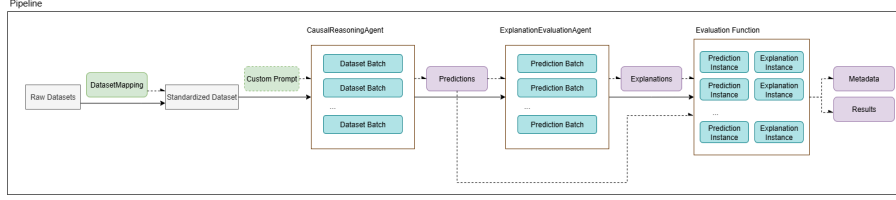


Figure 1: End-to-end pipeline. Raw datasets are mapped to a standardized schema via a user-specified *DatasetMapping*. The *CausalReasoningAgent* produces instance-level predictions and explanations. The *ExplanationEvaluationAgent* judges explanation validity and confidence. The *Evaluation* function aggregates dataset-level metrics and metadata.

represented as:

$$s_i = (\text{id}_i, \text{context}_i, \text{qtype}_i, \mathcal{C}_i, y_i, e_i^*),$$

where:

- **id** – a unique identifier for the instance,
- **context** – the background text required for answering the causal question. *Importantly, the causal question is always appended at the end of the context*, ensuring that datasets without explicit question columns can still be represented consistently,
- **question_type** – the type of causal relation being queried (*cause* or *effect*),
- **choices** – the finite set of answer options,
- **label** – the ground-truth answer (element of **choices**),
- **explanation** – the ground-truth explanation, providing semantic and causal justification.

3.2 Stage I: Dataset Mapping and Standardization

The *DatasetMapping* is implemented as a dictionary-like object specifying raw column names that realize the standardized fields in Table 1. Formally, $\phi : \mathcal{X}_{\text{raw}} \rightarrow \mathcal{X}_{\text{std}}$ maps raw columns to **context**, **question_type**, **choices**, **label**, and **explanation**. The mapping is validated for completeness and type consistency (e.g., **label** $\in$ **choices**), after which $\mathcal{D}^{\text{std}}$ is materialized and persisted. Persisted standardized datasets can be reused across models and runs, enabling controlled comparisons.

Table 1: Standardized schema produced by the *DatasetMapping*.

Field	Description
<code>id</code>	Unique identifier for the instance.
<code>context</code>	Background information required to answer the causal question. The explicit question is always appended at the end of the context, rather than stored as a separate field. This provides flexibility for datasets where a distinct question column is not natural.
<code>question_type</code>	Type of query: <i>cause</i> or <i>effect</i> .
<code>choices</code>	Discrete set of permissible answers for the instance.
<code>label</code>	Ground-truth answer (must be in <code>choices</code>).
<code>explanation</code>	Reference reasoning supporting the ground-truth answer.

3.3 Stage II: Causal Reasoning Agent

The *CausalReasoningAgent* applies an LLM to each $s_i \in \mathcal{D}^{\text{std}}$ to produce an answer and an accompanying model-generated explanation. In practice, instances are processed in batches of size B and parallelized across workers to maximize throughput. For each instance i , the agent returns a *Prediction Record*

$$p_i = (i, \hat{y}_i, y_i, \mathbf{1}[\hat{y}_i = y_i], v_i, \hat{e}_i, e_i^*, c_i^{\text{conf}}, \text{qtype}_i, \kappa_i^{\text{pred}}),$$

where $\hat{y}_i$ is the predicted answer, $v_i \in \{0, 1\}$ indicates whether $\hat{y}_i \in \mathcal{C}_i$ (guarding against hallucinated options), $\hat{e}_i$ is the model’s explanation, c_i^{conf} is the model-reported confidence, and κ_i^{pred} is the token cost usage for instance i .

Custom Prompting. A key feature of this agent is its support for *custom prompts*. In addition to the default system prompt (which instructs the model to evaluate causal-effect relations based on the standardized fields), users may supply tailored prompts to test the impact of different instruction styles on prediction accuracy and explanation quality. This design allows systematic experimentation with prompt engineering strategies.

Prompting and Guardrails. Prompts include the standardized fields and explicitly constrain answers to the enumerated `choices`. The validity flag v_i ensures downstream metrics can distinguish between incorrect but valid choices and out-of-support predictions.

3.4 Stage III: Explanation Evaluation Agent

The *ExplanationEvaluationAgent* judges whether $\hat{e}_i$ (from Stage II) is semantically aligned with and causally consistent relative to e_i^* . To control cost, we adopt a pragmatic policy: if $\hat{y}_i \neq y_i$, we omit explanation evaluation for instance

i (the justification of an incorrect answer is not required for our primary metrics). The agent processes predictions in batches (with parallelism) and returns *Explanation Records*

$$g_i = (i, \mathbf{1}[\text{explanation correct}], c_i^{\text{expl}}, \hat{e}_i, e_i^*, \kappa_i^{\text{expl}}),$$

where $c_i^{\text{expl}} \in [0, 1]$ is an evaluator-reported confidence and κ_i^{expl} the evaluation cost for i .¹

Evaluation Criteria. The evaluator is instructed to (i) assess semantic similarity to e_i^* (content overlap, entailment) and (ii) verify causal validity (directionality, necessary/sufficient relations as applicable to the question type). These criteria are expressed via structured rubrics in the prompt.

Incorrect Predictions. For instances where $\hat{y}_i \neq y_i$, the explanation is by definition marked as *incorrect*. In such cases, the system assigns an explanation correctness flag of 0 and sets the confidence to 1.0 deterministically. This convention avoids unnecessary model calls and ensures that explanation accuracy statistics properly reflect the fact that explanations for wrong answers cannot be considered valid.

3.5 Stage IV: Evaluation and Reporting

Given $\{p_i\}$ and $\{g_i\}$, we compute dataset-level metrics and produce two artifacts: *Results* (summary metrics) and *Metadata* (run descriptors). Let $\mathcal{I}$ be the index set of all instances, and $\mathcal{J} \subseteq \mathcal{I}$ the subset for which explanations were evaluated (i.e., $\hat{y}_i = y_i$).

Prediction Metrics.

$$\text{Total Predictions} = |\mathcal{I}|,$$

$$\text{Correct Predictions} = \sum_{i \in \mathcal{I}} \mathbf{1}[\hat{y}_i = y_i],$$

$$\text{Valid-Choice Rate} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} v_i,$$

$$\text{Prediction Accuracy} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbf{1}[\hat{y}_i = y_i],$$

$$\text{Prediction Cost} = \sum_{i \in \mathcal{I}} \kappa_i^{\text{pred}},$$

$$\text{Prediction Time} = \text{wall-clock elapsed for Stage II.}$$

¹The evaluator’s confidence is the LLM’s self-assessment; we aggregate it as a descriptive statistic rather than a calibrated probability.

Explanation Metrics.

Total Explanations Evaluated = $|\mathcal{J}|$,

$$\text{Explanation Accuracy} = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} \mathbf{1}[\text{explanation correct}],$$

$$\text{Avg. Explanation Confidence} = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} c_i^{\text{expl}},$$

$$\text{Explanation Cost} = \sum_{i \in \mathcal{J}} \kappa_i^{\text{expl}},$$

Explanation Time = wall-clock elapsed for Stage III.

Result Artifact. For each dataset we serialize:

- **Results:** dataset name; prediction model identifier; total, correct, and valid-choice counts; prediction accuracy; prediction cost and time; explanation model identifier; total and correct explanations; explanation accuracy; average explanation confidence; explanation cost and time.
- **Metadata:** dataset name; total cost (predictions/explanations), total execution time and model names/versions to ensure reproducibility.

In addition, raw **Predictions** $\{p_i\}$ and **Explanations** $\{g_i\}$ are persisted to enable fine-grained post-hoc analysis.

3.6 Parallelism and Cost Control

Batching and Parallelism. The system uses batch size B and worker concurrency W to process instances in parallel in both Stage II and Stage III. We log per-batch latency and throughput to characterize scaling behavior.

Cost Controls. We (i) enforce choice-constrained decoding via prompting and post-hoc validation (v_i), (ii) skip explanation evaluation when $\hat{y}_i \neq y_i$, and (iii) record per-instance token usage costs ($\kappa_i^{\text{pred}}, \kappa_i^{\text{expl}}$) for budget accounting.

3.7 Algorithmic Summary

1. **Standardize:** Apply *DatasetMapping* ϕ to $\mathcal{D}^{\text{raw}}$ to obtain $\mathcal{D}^{\text{std}}$; persist.
2. **Predict:** For each batch in $\mathcal{D}^{\text{std}}$, run the *CausalReasoningAgent* to produce $\{p_i\}$; persist predictions and timing/cost logs.
3. **Evaluate Explanations:** For instances with $\hat{y}_i = y_i$, run the *ExplanationEvaluationAgent* to produce $\{g_i\}$; persist with timing/cost logs.
4. **Aggregate:** Compute dataset-level metrics (Sec. 3.5) and serialize **Results** and **Metadata**.

Outputs. All artifacts—**Predictions**, **Explanations**, **Results**, and **Meta-data**—are stored under an experiment run directory for downstream analysis and visualization.

4 Experiments and Results

This section describes the experimental setup used to evaluate large language models (LLMs) on causal reasoning tasks, as well as the main results. We outline the datasets, experimental procedure, models and prompts, baselines, and the metrics recorded. Finally, we present aggregated results across datasets and discuss key observations.

4.1 Datasets

We evaluate four datasets spanning code-grounded, knowledge-grounded, mathematical, and textual causal reasoning:

1. **e-CARE**: a commonsense causal dataset with entries of the form {premise, ask-for $\in$ {cause, effect}, hypothesis1, hypothesis2, label $\in$ {0,1}, conceptual_explanation}. Each example is cast as a binary multiple-choice task with hypothesis1 and hypothesis2 as options, and the conceptual explanation as reference.
2. **CausalBench-code (code)**: executable Python-like causal scenarios with question types (cause-to-effect, effect-to-cause, with/without intervention), ground-truth answers, and short reference explanations.
3. **CausalBench-math (math)**: mathematically framed causal scenarios with the same four question families, ground-truth answers, and explanations.
4. **CausalBench-text (text)**: natural-language causal scenarios, often with probabilities or qualitative relations, again with ground-truth answers and short explanations.

4.2 Experimental Procedure

Each dataset is evaluated using the pipeline described in Section 3, where every instance undergoes prediction followed by explanation evaluation. Both stages use the same LLM: first as a solver that generates an answer with a justification, and then as a judge that assesses explanation quality.

Due to cost constraints, we evaluate exactly 100 instances per dataset in a single run, taken in dataset order without filtering or augmentation. Execution time and API usage cost are recorded for both stages. All results are persisted as per-item JSON records, while dataset-level metrics are aggregated at the end of each run.

4.3 Models and Prompting

All experiments use `gpt-4o-mini` for both reasoning and judging.

Prediction prompts. These instruct the model to answer the causal question and to provide a justification. For each instance i , the model outputs a predicted answer $\hat{y}_i$ together with a free-form explanation $\hat{e}_i$.

Evaluation prompts. These instruct the model-judge to compare the predicted explanation $\hat{e}_i$ against the reference explanation e_i^* , returning (i) a binary judgement $\mathbf{1}[\text{explanation correct}] \in \{0, 1\}$ and (ii) a confidence score $c_i^{\text{expl}} \in [0, 1]$.

Default decoding parameters are used, with no temperature or sampling overrides. Justifications are concise rationales without chain-of-thought prompting. No retrieval or external tools are employed. All calls are stateless.

4.4 Baselines

Two conceptual baselines provide context for performance:

- **Random guess:** 50% expected accuracy for balanced binary-choice tasks.
- **Majority class:** always selecting the most frequent label in the dataset.

We do not run models for these baselines; they serve as interpretive references.

4.5 Metrics

The framework records both prediction and explanation-level metrics as formalized in Section 3.5. So once again, for each dataset we report aggregate statistics such as prediction accuracy, valid-choice rate, execution time, and API cost. For explanations, we measure alignment accuracy, average evaluator confidence, execution time, and cost.

4.6 Results

We evaluate 100 instances per dataset. Aggregate results are summarized in Table 2 and visualized in Figures 2–5.

Table 2: Aggregate results (100 examples per dataset; single run). Costs are reported separately for prediction and explanation.

Dataset	Prediction Accuracy	Explanation Accuracy	Avg. Conf.	Cost (USD)
Code	0.82	0.49	0.86	0.0109 + 0.0058
e-CARE	0.88	0.05	0.71	0.0200 + 0.0115
Math	0.70	0.49	0.81	0.0113 + 0.0060
Text	0.70	0.32	0.87	0.0419 + 0.0226

Prediction accuracy substantially exceeds random guessing across all datasets, with the strongest results on `code` and `e-CARE`. However, explanation accuracy is

consistently lower, with the largest gap on **e-CARE**, where only 5% of generated justifications were judged as semantically aligned with the reference explanations. This mismatch highlights a central challenge: correct answers do not necessarily imply correct or faithful explanations. In contrast, valid-choice ratios remain close to 100% across all datasets, indicating that the model reliably restricts itself to the provided answer options.

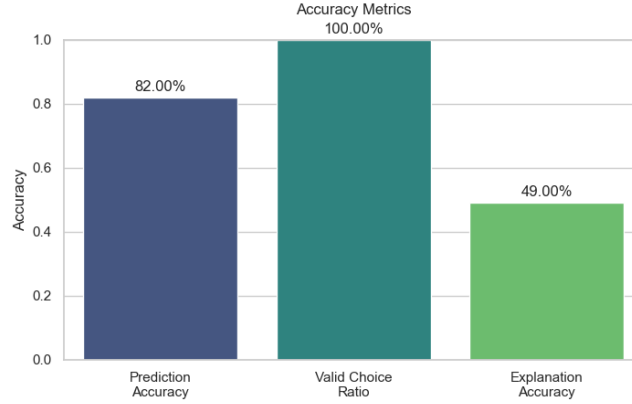


Figure 2: Accuracy metrics for the **code** dataset.

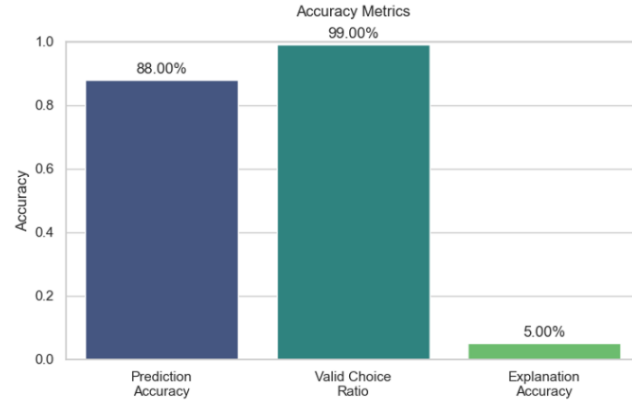


Figure 3: Accuracy metrics for the **e-CARE** dataset.

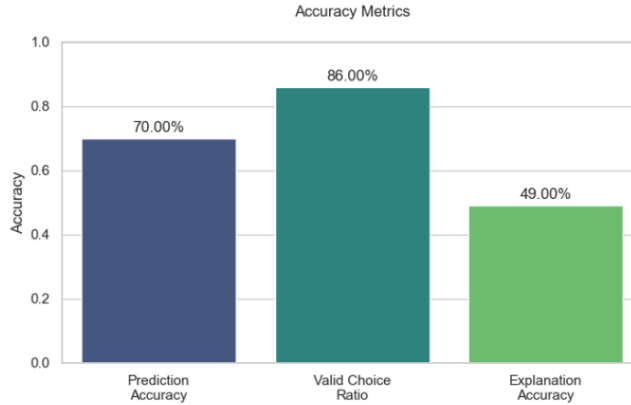


Figure 4: Accuracy metrics for the **math** dataset.

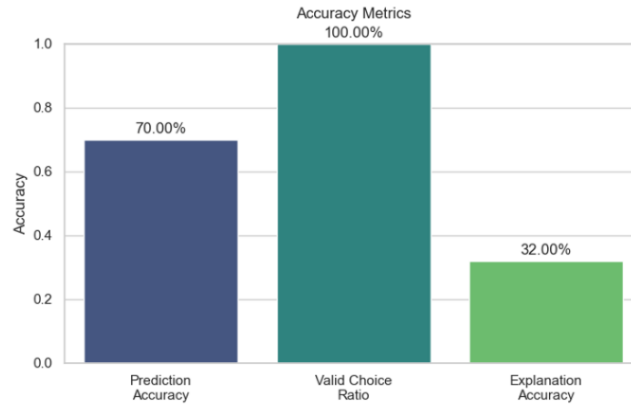


Figure 5: Accuracy metrics for the **text** dataset.

The visualizations reinforce the numerical findings: prediction accuracies are consistently higher than explanation accuracies, with explanation performance varying widely between datasets. While **code** and **math** achieve nearly 50% explanation accuracy, **text** lags behind at 32%, and **e-CARE** shows an especially stark contrast between high prediction accuracy (88%) and very poor explanation accuracy (5%). These results suggest that short, low-cost justifications often fail to align with conceptual explanations, even when answers are correct.

5 Qualitative Examples

To illustrate the behavior of the framework beyond aggregate metrics, we present qualitative examples from the **text** dataset. These cases highlight both the

strengths and weaknesses of LLM-based causal reasoning, as well as the importance of evaluating explanations separately from predictions.

5.1 Successful Example

Prediction ID 5 asked whether reducing tar deposits among smokers decreases lung cancer risk. The model correctly answered *Yes* and produced the following explanation:

“Reducing tar deposits among smokers lowers the probability of lung cancer, as evidenced by the lower lung cancer rates associated with lower tar deposits.”

This explanation was semantically aligned with the ground-truth and was judged correct by the evaluation agent. The case illustrates the framework’s strength in capturing straightforward causal relations, particularly when probabilities are clearly structured and the causal mechanism is intuitive.

5.2 Failure Example

Prediction ID 1 asked whether promoting non-smoking would reduce lung cancer incidence. The model again answered *Yes*, which matched the ground-truth label. However, its explanation diverged sharply:

“Promoting non-smoking reduces the number of smokers, which decreases the probability of high tar deposits and subsequently lowers the lung cancer rates associated with both smoking and tar deposits.”

In contrast, the ground-truth explanation emphasized that promoting non-smoking paradoxically *increases* overall lung cancer incidence in this dataset, due to higher baseline risk among nonsmokers. The evaluation agent therefore marked the explanation as incorrect. This example underscores that even when predictions are correct, explanations may be shallow or misleading, particularly in counterintuitive or dataset-specific scenarios.

5.3 Interpretation

These examples demonstrate the value of our dual-agent framework. In straightforward cases, the system reliably captures causal dependencies and provides valid explanations. In more complex or counterintuitive cases, however, the evaluation agent exposes when explanations fail to reflect the underlying causal structure. This distinction between correctness of predictions and validity of explanations highlights why both dimensions must be evaluated for a complete assessment of LLM causal reasoning.

6 Limitations

While the proposed framework provides a systematic and extensible approach to evaluating causal reasoning in LLMs, it also inherits several important limitations.

First, although the framework standardizes diverse causal datasets into a unified schema, its generality across highly specialized domains (e.g., economics, biology, or medicine) remains limited without additional domain-specific prompting or fine-tuning. Domain adaptation may be necessary for robust evaluations in these settings.

Second, the framework is inherently sensitive to the design of prompt templates. Small differences in phrasing between datasets and task descriptions can influence both predictions and explanations, which may obscure whether performance differences reflect genuine reasoning ability or prompt artifacts.

Third, due to budget constraints, all experiments were conducted using `gpt-4o-mini`, a relatively lightweight and inexpensive model. While this allowed us to demonstrate the feasibility of the pipeline, the reported results should not be interpreted as upper bounds on performance. Stronger models may yield significantly higher prediction and explanation accuracy. In our work, the primary contribution lies in the design of the evaluation framework itself, rather than the absolute performance numbers obtained.

Finally, as with all LLM-based approaches, reproducibility is constrained by the nondeterminism of generative models and the dependence on external API providers. Although our pipeline records per-item outputs and aggregated metrics, minor numerical differences may arise when experiments are rerun.

These limitations highlight important directions for future work: incorporating stronger LLMs, exploring domain-specific adaptations, and reducing prompt sensitivity, all of which can improve the robustness and generality of causal reasoning evaluation.

7 Conclusion

In this work, we introduced a systematic framework for evaluating the causal reasoning abilities of large language models (LLMs). Unlike prior efforts that often target narrow, domain-specific benchmarks, our approach unifies datasets from code, mathematics, textual scenarios, and commonsense causal inference into a standardized schema. This design enables consistent cross-domain evaluation and provides a foundation for reproducibility and extensibility.

The framework adopts a dual-agent pipeline: a reasoning agent that produces predictions with justifications, and an evaluation agent that assesses whether explanations are semantically aligned and causally faithful. This two-dimensional perspective—examining both correctness of answers and validity of explanations—offers a more nuanced assessment of causal reasoning than prediction accuracy alone.

Experimental results demonstrate that LLMs reliably exceed random baselines in prediction accuracy, with particularly strong gains in structured domains such as code and commonsense reasoning. However, explanation accuracy remains significantly lower across all datasets, in some cases dropping below 10%. Moreover, models often displayed high confidence in explanations that were ultimately judged incorrect. These findings highlight that accurate answers do not necessarily reflect robust causal reasoning, and that explanation evaluation is essential for fair benchmarking.

By combining dataset standardization, modular evaluation, and open-source implementation, our framework provides a transparent and extensible foundation for future research. We view this as a step toward more reliable benchmarking of LLMs’ causal reasoning capabilities, emphasizing not only what models predict, but why they predict it.

8 Future Work

Several promising directions remain for extending this framework.

First, incorporating domain-specific datasets—such as those from biology, economics, or the social sciences—would provide a stronger test of whether LLMs generalize causal reasoning beyond the four domains studied here. Domain adaptation and specialized prompting could help address the current limitations in highly specialized fields.

Second, the evaluation agent itself could be improved. Instead of relying solely on a single LLM’s judgment, future work may explore embedding-based semantic similarity, causal graph alignment, or hybrid scoring methods that combine automatic metrics with LLM-based assessments. Such approaches could increase both reliability and interpretability of explanation evaluation.

Third, the framework could be scaled to larger datasets and a broader range of model families. Running experiments with stronger models, as well as lightweight variants, would allow for statistical robustness and fairer comparisons across architectures and budget levels.

Finally, extending the framework to support hybrid evaluation pipelines where automatic scoring provides baseline consistency and LLM-based agents contribute deeper qualitative judgments—may help balance cost, scalability, and insight.

Taken together, these extensions would improve the breadth, reliability, and fairness of causal reasoning benchmarks, moving the field closer to systematic and trustworthy evaluation of LLM reasoning.

A Reproducibility

All code, standardized datasets, prompts, and experimental logs are available in our public repository:

<https://github.com/Brauo0ue/causal-graph-llm-evaluator>

The repository contains a detailed **README** with step-by-step instructions to reproduce every experiment described in this paper. This includes environment setup, dataset preparation, pipeline execution, and evaluation scripts. Users can also adjust model choices, prompts, and dataset sizes, enabling flexible experimentation beyond the fixed runs reported here.

Our design emphasizes transparency and openness: the repository includes per-item JSON outputs, aggregated metrics, and logging utilities, ensuring that results are fully auditable and extendable.

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